

High-frequency parameter updates in digital twins: Report on a 2-week challenge test for fully autonomous adaptive operation of an effluent water recovery system.

Pooria Ghorbani Bam*, Subrata Mukherjee, Alexander M. Melin**, Brian J. Tarroja*, Sally Ghanem**, Kris Villez**, Diego Rosso*,⁺**

* University of California, Irvine, Water-Energy Nexus Center, 844G Engineering Tower, Irvine, CA 92697-2175 (USA); +: Tel: +1-310-430-1687; E-mail: bidui@uci.edu

** Oak Ridge National Laboratory, Electrification and Energy Infrastructure Division, 1 Bethel Valley Road, TN 37920 (USA)

Introduction

Digital twins (DTs) that can predict the transient behaviour of a water treatment plant combined with buffer capacity, enable reductions in specific energy consumption (SEC) in response to time-of-use (TOU) electricity costs. The accuracy of a DT depends on the richness of its training data (Conejos Fuertes et al. 2020). However, process changes cannot easily be tested in production-scale facilities due to water-quality compliance and output requirements. To address this, UCI developed a heavily instrumented closed-circuit reverse osmosis (CCRO) physical twin of the OCWD plant. Experiments on the physical twin over a wide range of process parameters generate the rich dataset needed to train the DT. The trained DT can then use sparse plant-scale data to predict how process changes affect water quality, energy use, membrane fouling, etc.

In our research, we developed a method to train the DT using a rich experimental data set collected from the physical twin. Then we use the DT to find a time series of process parameters that minimize the SEC over a seven-day interval based on a TOU electricity cost schedule. To simplify the exposition in this work, factors such as water quality, membrane lifetime, and brine disposal were excluded from the optimization but can be incorporated in future cost-function constraints. Even without TOU pricing, optimal CCRO setpoints are not self-evident; this framework identifies them and is broadly applicable to other treatment systems.

Methodological Approach

Two parallel DTs were developed: (1) a machine-learning (ML) model and (2) a SUMO simulation via Dynamizu. As shown in Figure 1, the DT is updated hourly using the previous seven days of data (estimation horizon) and then optimizes hourly flow rate setpoints for the next seven days (control and forecasting horizon). The resulting 504 setpoints minimize expected operational cost under the given electricity tariff while satisfying safety and operational constraints. The optimized first-hour setpoints for permeate, concentrate, and recirculation flows are written to the pilot PLC.

The ML-based DT applies a sparsity-inducing LASSO algorithm (Tibshirani 2011; Fonti and Belitser 2017) relating steady-state power demand to three controlled flow rates. A cubic polynomial basis was used leading to a maximum of 19 polynomial coefficients, of which many are set to zero during model calibration. The model calibration, including selection of non-zero coefficients is repeated on an hourly basis.

The semi-autonomous DT ran for two weeks, storing pilot data in the cloud and updating the calibration file every hour using a rolling seven-day (168-hour) dataset.

The second DT, built in SUMO, estimates power consumption and permeate flow. Pilot experiments systematically varied feed flow, recovery, and recirculation ratios using Latin hypercube sampling to produce 64 safe setpoint combinations within the convex operational region defined by the pilot manual. Calibration minimized the difference between measured and predicted power by adjusting pump efficiencies. An autonomous control experiment with this DT is under way at the time of writing.

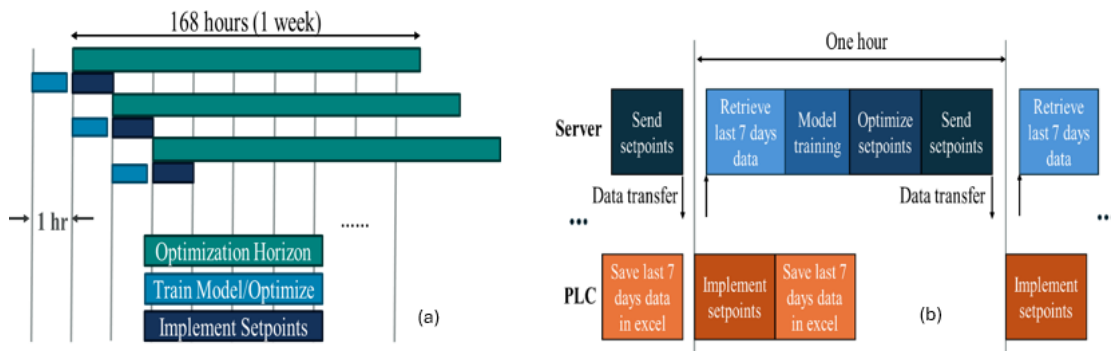


Figure 1. Model-predictive control scheme for DT-driven optimization; (a) overall loop, (b) hourly cycle.

Results and Discussion

The ML-based DT framework was implemented through a two-stage surrogate modelling and optimization pipeline. In the first stage, the surrogate model was retrained hourly as new data became available, predicting power consumption under varying flow rate setpoints. The optimization routine then minimized cumulative electricity cost over a 168-hour horizon under physical and safety constraints. In the second stage, only the first hour's optimized setpoints were applied to the system, ensuring receding-horizon control. To maintain robustness and hardware safety, optimized setpoints were smoothed through controlled perturbation and rounding to limit abrupt actuation. Model interpretability and temporal stability were evaluated by tracking regression coefficients over 332 hours (~two weeks). Figure 2 shows the normalized temporal evolution of these coefficients within $[-1, 1]$ for visual comparability. First-order terms, concentrate, permeate, and recirculation flow rates, remained persistently active, indicating convergence to a stable surrogate structure. In contrast, higher-order and interaction terms appeared intermittently, reflecting transient structural exploration followed by sparsity regularization as data accumulated.

Figure 3(a) illustrates a representative cost comparison between nominal and optimized operation at the 49th hour (after 2 days of real time deployment). The nominal operation corresponding to steady-state flow rates would have incurred an energy cost of approximately \$203.78, whereas the ML based DT-optimized schedule reduces this to \$122.83, yielding a cost savings of 39.97%. Averaged across the full 332-hour window, this DT framework achieved a consistent average cost reduction of 38.33%. Figure 3(b) further illustrates how the predicted power profile adapts to the

real-time tariff structure. As expected, the model learns to suppress power usage during high-tariff periods while shifting operational load to lower-tariff hours, validating the economic responsiveness of the surrogate-driven controller. Since only the first setpoints of each 168-hour optimization are implemented, Figure 4(a) plots the initial optimized permeate flow rate setpoints selected for implementation over the full 2-week period. The plot demonstrates that the optimization systematically allocates higher flow rates during low tariff periods and reduces them during tariff surges. Figure 4(b) shows the absolute hourly change in permeate flow rate across consecutive setpoints. The magnitude of changes remains minimal (typically $< 0.2 \text{ gpm}$) during periods of tariff flatness, while sharp transitions in tariff values result in corresponding adjustments to the setpoints. This behaviour confirms that the optimization framework selectively modifies control decisions only when economically justified, promoting stability and smooth transitions during constant-tariff intervals.

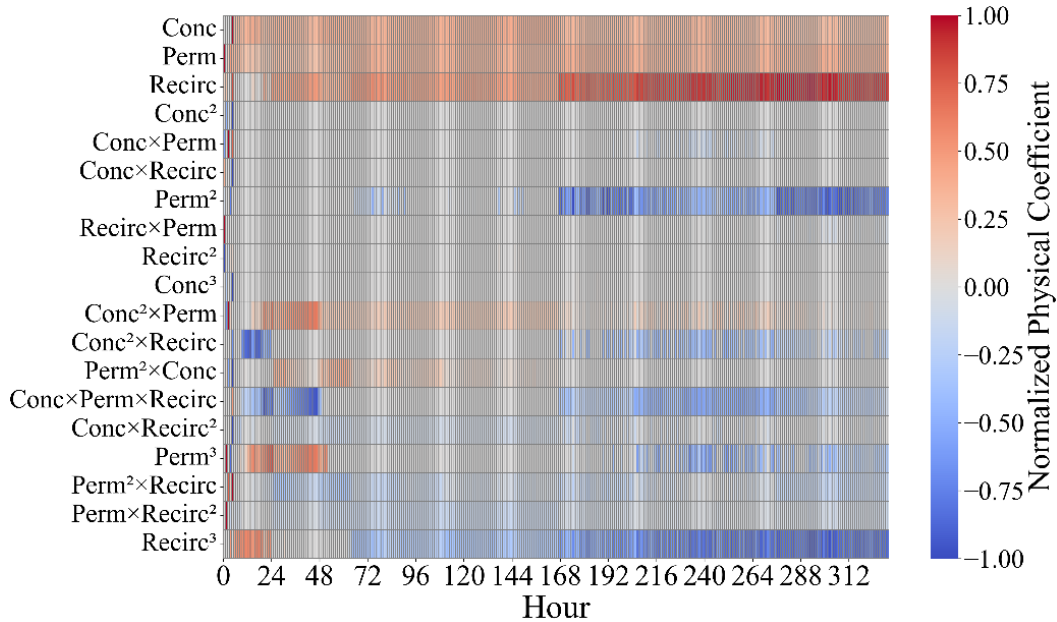


Figure 2: Temporal evolution of normalized regression coefficients over the 2-week.

Conclusions

The results demonstrate that the developed DT framework can autonomously adapt pilot operation to dynamic electricity tariffs and achieve substantial energy savings. The implementation achieved an average 38.33% reduction in operating cost relative to nominal steady-state conditions while maintaining stable control performance and interpretable model structure. These findings confirm that high-frequency parameter updates enable data-driven adaptability without compromising operational robustness, providing a scalable pathway toward fully autonomous, energy-responsive water reuse systems.

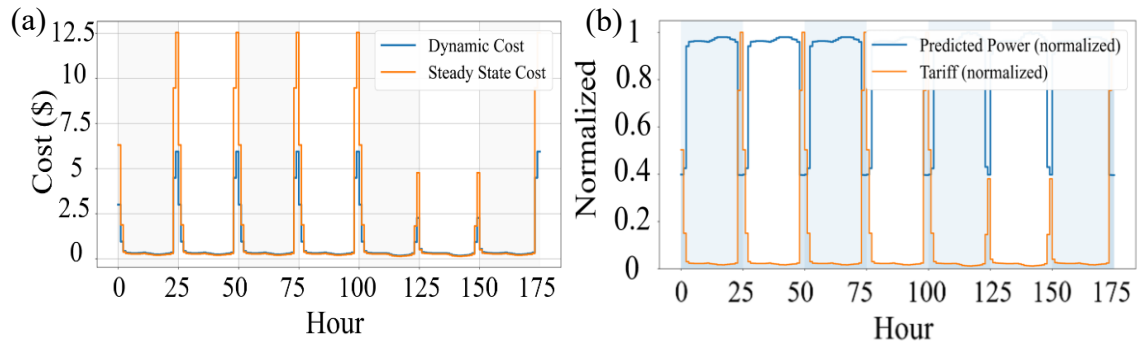


Figure 3: (a) Comparison of electricity cost under nominal steady-state operation and optimized control at the 49th hour of deployment, (b) Predicted power profile overlaid with the tariff structure, demonstrating cost-aware adaptation.

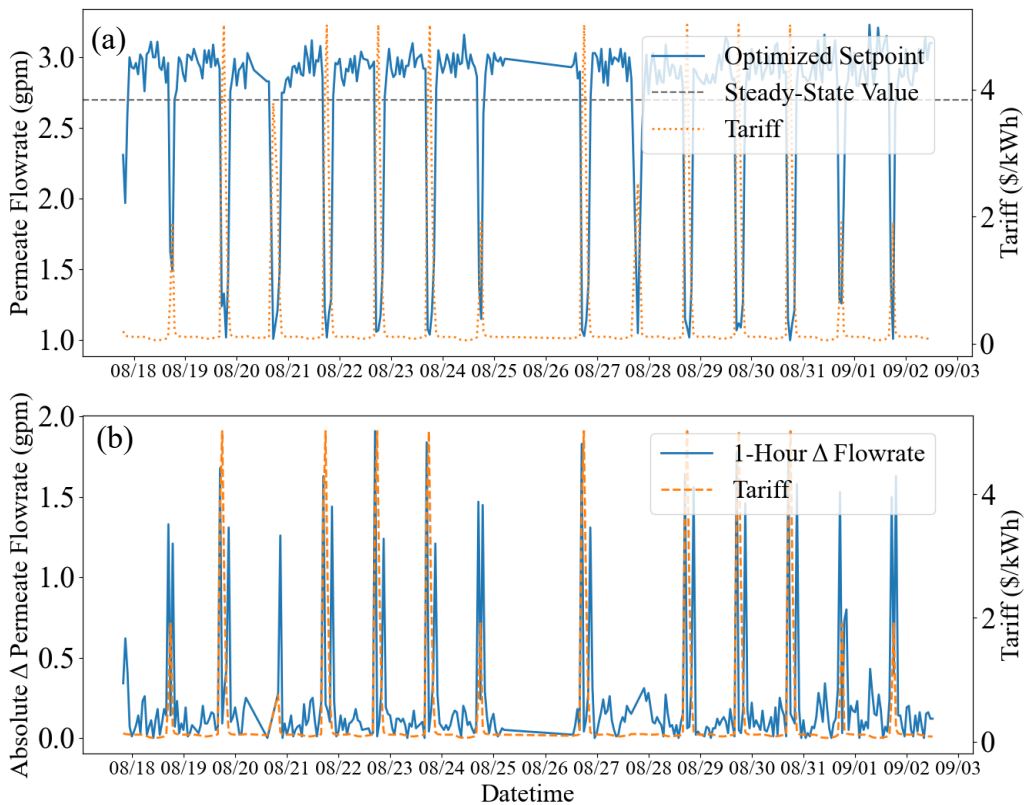


Figure 4: (a) Hourly DT-optimized permeate flow rate setpoints deployed to the physical pilot, overlaid with the tariff signal over the 2-week period, (b) Absolute change in permeate flow rate between consecutive hours, illustrating response dynamics relative to the tariff.

References

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